



Experiment-10

Student Name: Pranav Chauhan
Branch: BE CSE (cyber security)
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Subject Name: Computer Networks

UID: 24BCY70004
Section/Group: 24AIT_KRG-G2
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1. Aim/Overview of the practical:

To capture and analyze network packets using Wireshark by monitoring network traffic and examining protocols such as **ICMP, ARP, DNS, and TCP**.

2. Software/Tools Required:

- Wireshark
- Command Prompt (CMD)
- Ethernet / Wi-Fi / Mobile Hotspot
- Windows Operating System
- Network interface such as Ethernet or Wi-Fi adapter

3. Theory:

A) Wireshark

Wireshark is a network packet analyzer used to capture and analyze network traffic. It can inspect packets passing through a selected network interface.

B) Network Interface

A network interface is the connection through which a computer communicates with a network, such as Ethernet or Wi-Fi. The correct active interface must be selected before starting packet capture.

C) Promiscuous and Monitor Mode

In Promiscuous Mode, the network interface can capture packets other than those addressed specifically to the device. Monitor Mode is used with wireless interfaces to capture wireless frames in the surrounding network environment.

D) Packet Capture

Wireshark captures network packets through supporting tools such as libpcap/Npcap. The captured packets can then be filtered and analyzed according to different protocols such as ICMP, ARP, DNS, and TCP.

E) Packet Analysis

Filters can be applied in Wireshark to identify specific packets. For example, ICMP packets can be filtered using `icmp && ip.addr == 8.8.8.8`, ARP using `arp`, DNS using `dns.qry.name == "example.com"`, and TCP using `tcp.port == 80`.

4. Steps for Experiment/Practical:

- 1) Connect the laptop to the Internet using Ethernet, Wi-Fi, or mobile hotspot.
- 2) Open Wireshark and select the active network interface.
- 3) Start packet capture by double-clicking the selected interface.
- 4) For ICMP capture, open CMD and run `ping 8.8.8.8`.
- 5) Stop the capture and apply the filter `icmp && ip.addr == 8.8.8.8`.
- 6) Identify the Echo Request and corresponding Echo Reply packets.
- 7) For ARP capture, run `ipconfig` and note the Default Gateway IP.
- 8) Open CMD as Administrator and run `arp -d *`.
- 9) Start Wireshark capture and run `ping <Default-Gateway-IP>`.
- 10) Stop the capture and apply the filter `arp`.
- 11) Identify the ARP Request and corresponding ARP Reply, and analyze their IP/MAC addresses.
- 12) For DNS capture, start Wireshark and open CMD.
- 13) Run `nslookup example.com`.
- 14) Stop the capture and apply the filter `dns.qry.name == "example.com"`.
- 15) Identify the A Query and its Response and inspect the DNS and UDP fields.
- 16) For TCP capture, start Wireshark and open CMD.
- 17) Run `curl http://example.com`.
- 18) Stop the capture and apply the filter `tcp.port == 80`.
- 19) Identify the TCP connection and observe the [SYN] → [SYN, ACK] → [ACK] sequence.
- 20) Analyze the source, destination, ports, sequence/acknowledgment numbers, and TCP flags.

5. Output:

dns.qry.name == "example.com"

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	192.168.13.128	9.9.9.9	DNS	71	Standard query query
2	0.000158377	192.168.13.128	9.9.9.9	DNS	71	Standard query response
3	0.084955544	9.9.9.9	192.168.13.128	DNS	103	Standard query response
4	0.084956123	9.9.9.9	192.168.13.128	DNS	127	Standard query response

> Frame 1: Packet, 71 bytes on wire (568 bits), 71 bytes captured (568 bits) on interface
 > Ethernet II, Src: VMware_26:f5:95 (00:0c:29:26:f5:95), Dst: VMware_e6:a4:64 (00:50:56:e6:a4:64)
 > Internet Protocol Version 4, Src: 192.168.13.128, Dst: 9.9.9.9
 > User Datagram Protocol, Src Port: 54624, Dst Port: 53
 > Domain Name System (query)

icmp && ip.addr == 8.8.8.8

No.	Time	Source	Destination	Protocol	Length	Info
33	116.317806381	192.168.13.128	8.8.8.8	ICMP	98	Echo (ping) request
36	116.335740860	8.8.8.8	192.168.13.128	ICMP	98	Echo (ping) response
37	117.320373544	192.168.13.128	8.8.8.8	ICMP	98	Echo (ping) request
38	117.356456858	8.8.8.8	192.168.13.128	ICMP	98	Echo (ping) response
39	118.321786171	192.168.13.128	8.8.8.8	ICMP	98	Echo (ping) request
40	118.363704219	8.8.8.8	192.168.13.128	ICMP	98	Echo (ping) response
41	119.323354625	192.168.13.128	8.8.8.8	ICMP	98	Echo (ping) request
42	119.342065397	8.8.8.8	192.168.13.128	ICMP	98	Echo (ping) response

> Frame 41: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface
 > Ethernet II, Src: VMware_26:f5:95 (00:0c:29:26:f5:95), Dst: VMware_e6:a4:64 (00:50:56:e6:a4:64)
 > Internet Protocol Version 4, Src: 192.168.13.128, Dst: 8.8.8.8
 > Internet Control Message Protocol



6. Result

The network packets were successfully **captured and analyzed using Wireshark**. ICMP, ARP, DNS, and TCP packets were identified using appropriate Wireshark filters, and their relevant fields and communication sequences were examined.

7. Learning Outcomes

- Understand the basic working of **Wireshark** as a network packet analyzer.
- Learn to capture network packets using an active network interface.
- Identify and analyze **ICMP, ARP, DNS, and TCP** packets.
- Learn to apply Wireshark filters for specific network protocols.
- Analyze packet information such as **source, destination, protocol, ports, and packet fields**.